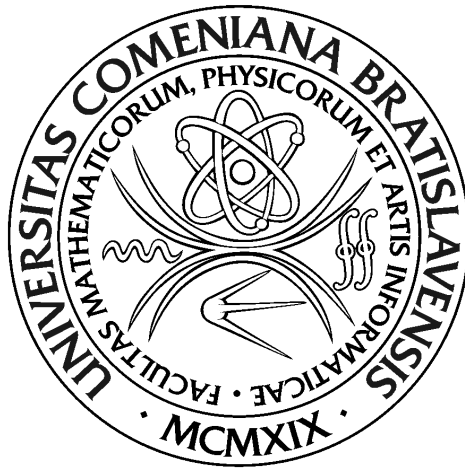


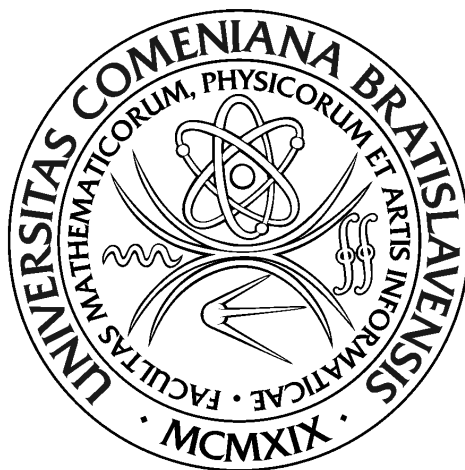
COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS



SHAPLEY-BASED CREDIT ASSIGNMENT FOR THE DREAMER ALGORITHM IN TERMINAL-REWARD TASKS

Master's Thesis

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Master's Thesis

Study program: Applied Informatics
Branch of study: Applied Informatics
Department: Department of Applied Informatics
Supervisor: Mgr. Marek Šuppa

I hereby declare that I have written this thesis by myself, only with the help of the referenced literature, under the careful supervision of my thesis advisor.

Bratislava, 2026

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Bc. Aleksandr Bukhtoiarov

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Abstract

One of the fundamental problems in reinforcement learning is credit assignment, that is, determining the contribution of individual actions to the final reward. This problem is particularly pronounced in tasks with terminal rewards, where the entire episode is evaluated by a single scalar signal and no intermediate feedback is available. In such settings, the algorithm receives no information about which actions contributed positively or negatively to the outcome, leading to high gradient variance and making learning difficult, especially for long action sequences.

The recently proposed method SCAR (Shapley Credit Assignment for More Efficient RLHF) formulates sequence generation as a cooperative game and uses Shapley values to decompose a terminal reward into contributions of individual tokens. In the context of RLHF, this makes it possible to replace a single sparse reward with a dense sequence of token-level rewards, which can be used for more stable and efficient fine-tuning of large language models.

In this thesis, the idea of Shapley-based credit assignment is transferred from RLHF to the Dreamer algorithm. Unlike RLHF, where reward evaluation requires a separate reward model trained on human preferences, Dreamer already includes a learned model of environment dynamics and reward. This allows counterfactual variants of trajectories to be evaluated using latent imagined rollouts, without additional interaction with the real environment.

The work focuses on text-based and text-procedural tasks in which an agent executes a sequence of discrete actions (such as text generation, memory operations, or module calls) and receives a reward only at the end of the episode. After an episode is completed, the terminal reward is distributed among individual actions using Shapley sampling. This is achieved by estimating the expected terminal reward for multiple modified variants of the same trajectory, obtained by replacing a subset of actions with a baseline action, and evaluating them through latent rollouts of the Dreamer environment model. The resulting sequence of action contributions is used as a dense reward signal for training both the policy and the dynamics model.

In the experimental part, Dreamer is compared under three training regimes: using only a terminal reward, using manual reward shaping, and using Shapley-based reward decomposition. The evaluation focuses on learning speed, training stability, and the

quality of the learned policy, as well as on the impact of the number of Shapley samples and the action segmentation strategy.

Keywords: reinforcement learning, world models, Dreamer, credit assignment, Shapley values, terminal reward, RLHF

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Terminology

Symbols

- $\mathcal{S}, \mathcal{A}$ – state and action spaces of an MDP.
- $\pi(a \mid s)$ – policy: a distribution over actions conditioned on a state.
- r_t, R – per-step reward and the (terminal) episode return.
- $\tau = (s_0, a_0, s_1, a_1, \dots, s_T)$ – trajectory.
- $\phi_i(v)$ – Shapley value of player i in cooperative game with value function v .
- z_t – latent state of a world model at time t .
- $a^\perp$ – baseline (no-op) action used to construct counterfactual coalitions.
- M – number of Monte Carlo Shapley samples per trajectory.

Abbreviations

- **RL** – Reinforcement Learning.
- **MDP** – Markov Decision Process.
- **POMDP** – Partially Observable Markov Decision Process.
- **TD** – Temporal Difference.
- **PPO** – Proximal Policy Optimization.
- **RLHF** – Reinforcement Learning from Human Feedback.
- **LLM** – Large Language Model.
- **SCAR** – Shapley Credit Assignment for More Efficient RLHF.
- **RSSM** – Recurrent State-Space Model (Dreamer’s world model).
- **HCA** – Hindsight Credit Assignment.
- **RUDDER** – Return Decomposition for Delayed Rewards.

- **KL** – Kullback–Leibler divergence.
- **GAE** – Generalized Advantage Estimation.

Motivation

Reinforcement learning (RL) systems learn through interaction: an agent acts in an environment and updates its policy from the rewards it observes. When rewards are dense, the credit assignment problem – attributing the eventual outcome to the individual actions that produced it – is largely solved by temporal-difference and policy-gradient methods that propagate signal step by step. The picture changes drastically when feedback is delivered only at the end of an episode. Long horizons, sparse signals, and noisy returns combine into very high gradient variance, slow convergence, and policies that learn from accident more than from understanding.

This work is motivated by a recent line of research that treats the trajectory as a cooperative game, with actions playing the roles of cooperating agents and the terminal reward acting as the value of the grand coalition. The SCAR method [8] demonstrated that distributing the terminal reward across tokens of a generated sequence using Shapley values produces a dense, fair, and theoretically grounded reward decomposition that significantly accelerates RLHF-style fine-tuning of large language models. Crucially, however, SCAR pays a high price for this signal: it relies on a separately trained reward model that approximates human preferences and that must be queried for many counterfactual sequences.

Model-based RL, and the Dreamer family of algorithms [3, 4] in particular, offers a complementary opportunity. Dreamer maintains a learned latent dynamics model and a learned reward predictor, both trained jointly with the policy. These components can be queried offline, in latent space, to evaluate counterfactual continuations of a trajectory *without further interaction with the real environment*. This makes Dreamer a natural host for Shapley-style credit assignment: the world model can supply exactly the kind of cheap counterfactual evaluations that the SCAR estimator demands, and it can do so for arbitrary action sequences rather than only for textual ones.

The aim of this thesis is to bring these two lines of work together. We propose a method that decomposes terminal rewards into per-action contributions using Monte Carlo Shapley sampling, with coalition values estimated through latent imagined roll-outs of a Dreamer-style world model. The dense per-action contributions are then fed back as a training signal for both the policy and the dynamics model. We focus on text-based and text-procedural tasks because they exhibit the long, discrete, terminally

rewarded structure where credit assignment is hardest, while remaining computationally tractable for prototype-scale experiments.

Chapter 1

Background

Chapter 2

Proposed method

Chapter 3

Implementation

Chapter 4

Experiments

Chapter 5

Conclusion

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